International Institute of Information Technology, Hyderabad

CS9.312-Introduction to Quantum Information and Computation

(Spring-2026)

Mid Semester Exam

Duration: 90 Minutes

Total Marks: 40

Question 1 [9 Marks]. Recall that any single–qubit density matrix ρ can be written in *Bloch form* as

$$\rho = \frac{1}{2}\left(\mathbb{I} + \vec{r}\cdot\vec{\sigma}\right),$$

where $\vec{\sigma} = (X, Y, Z)$ is the vector of Pauli matrices and $\vec{r} = (r_x, r_y, r_z) \in \mathbb{R}^3$ is called the *Bloch vector*. The Bloch vector satisfies $|\vec{r}| \leq 1$ and the set of all physical states corresponds to the unit ball in $\mathbb{R}^3$. Pure states lie on the surface $|\vec{r}| = 1$, while mixed states lie strictly inside the ball. Now let $0 \leq p \leq 1$ and define the channel

$$\mathcal{E}_p(\rho) = (1-p)\rho + \frac{p}{4}\left(\rho + X\rho X + Y\rho Y + Z\rho Z\right).$$

(a) **[4 Marks]** Show that

$$\mathcal{E}_p(\rho) = (1-p)\rho + p\frac{\mathbb{I}}{2}.$$

(b) **[2 Marks]** Using the Bloch representation, determine how the Bloch vector $\vec{r}$ transforms under $\mathcal{E}_p$.

(c) **[3 Marks]** Suppose the channel is applied repeatedly n times. What happens to the Bloch vector as $n \to \infty$?

Question 2 [9 Marks]. Construct and draw a quantum circuit to add two two-bit numbers x and y modulo 4. That is, the circuit should perform the transformation

$$|x, y\rangle \longrightarrow |x,\, x + y \bmod 4\rangle.$$

Verify the output of the circuit.

Question 3 [12 Marks]. The *majority function* on three bits is defined by

$$\mathrm{Maj}(a, b, c) = 1 \quad \text{iff at least two of } a, b, c \text{ are equal to 1.}$$

(a) **[2 Marks]** Write down the truth table of $\mathrm{Maj}(a, b, c)$.

(b) **[5 Marks]** Express $\mathrm{Maj}(a, b, c)$ as an algebraic formula using only XOR ($\oplus$) and AND (multiplication) operations. **[Hint:** You may use $x \vee y = x \oplus y \oplus xy$]

(c) **[5 Marks]** Consider the diagonal unitary

$$U_{\text{Maj}} |a, b, c\rangle_{ABC} = (-1)^{\text{Maj}(a,b,c)} |a, b, c\rangle_{ABC} \cdot$$

Construct and draw a quantum circuit for U_{Maj} using only CNOT and single–qubit gates.

Question 4 [10 Marks] Alice and Bob share the maximally entangled state $|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$. Alice secretly chooses one of the following two measurement strategies:

- **Strategy Z:** Measure in the computational basis $\{|0\rangle, |1\rangle\}$.
- **Strategy θ:** Measure in the rotated basis $\{|\psi_\theta\rangle, |\psi_\theta^\perp\rangle\}$, where $|\psi_\theta\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle$ and $|\psi_\theta^\perp\rangle = \sin\frac{\theta}{2}|0\rangle - \cos\frac{\theta}{2}|1\rangle$.

Alice does **not** reveal which strategy she used, and she does **not** send her measurement outcome to Bob.

1. **[4 Marks]** For each strategy, determine the ensemble of states prepared on Bob's side.
2. **[2 Marks]** Compute Bob's density matrix for each strategy.
3. **[2 Marks]** Can Bob determine which strategy Alice used by performing any measurement on his qubit?
4. **[2 Marks]** Now suppose Alice reveals which basis she used but not her measurement outcome. Does this change Bob's state?